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compact1, compact2, compact3

java.io

Class DataOutputStream

java.lang.Object
    java.io.OutputStream
        java.io.FilterOutputStream
            java.io.DataOutputStream

All Implemented Interfaces:

`Closeable`, `DataOutput`, `Flushable`, `AutoCloseable`

```
public class DataOutputStream
extends FilterOutputStream
implements DataOutput
```

A data output stream lets an application write primitive Java data types to an output stream in a portable way. An application can then use a data input stream to read the data back in.

Since:

JDK1.0

See Also:

`DataInputStream`

Field Summary

Fields

Modifier and Type	Field	Description
protected int	written	The number of bytes written to the data output stream so far.

Fields inherited from class java.io.FilterOutputStream

`out`

Constructor Summary

Constructors

Constructor	Description
`DataOutputStream(OutputStream out)`	Creates a new data output stream to write data to the specified underlying output stream.

Method Summary

Concrete Methods

Modifier and Type	Method	Description
`void`	`flush()`	Flushes this data output stream.
`int`	`size()`	Returns the current value of the counter `written`, the number of bytes written to this data output stream so far.
`void`	`write(byte[] b, int off, int len)`	Writes `len` bytes from the specified byte array starting at offset `off` to the underlying output stream.
`void`	`write(int b)`	Writes the specified byte (the low eight bits of the argument `b`) to the underlying output stream.
`void`	`writeBoolean(boolean v)`	Writes a `boolean` to the underlying output stream as a 1-byte value.

void	**writeByte**(int v)	Writes out a byte to the underlying output stream as a 1-byte value.
void	**writeBytes**(**String** s)	Writes out the string to the underlying output stream as a sequence of bytes.
void	**writeChar**(int v)	Writes a char to the underlying output stream as a 2-byte value, high byte first.
void	**writeChars**(**String** s)	Writes a string to the underlying output stream as a sequence of characters.
void	**writeDouble**(double v)	Converts the double argument to a long using the doubleToLongBits method in class Double, and then writes that long value to the underlying output stream as an 8-byte quantity, high byte first.
void	**writeFloat**(float v)	Converts the float argument to an int using the floatToIntBits method in class Float, and then writes that int value to the underlying output stream as a 4-byte quantity, high byte first.
void	**writeInt**(int v)	Writes an int to the underlying output stream as four bytes, high byte first.
void	**writeLong**(long v)	Writes a long to the underlying output stream as eight bytes, high byte first.
void	**writeShort**(int v)	Writes a short to the underlying output stream

		as two bytes, high byte first.
void	**writeUTF**(**String** str)	Writes a string to the underlying output stream using **modified UTF-8** encoding in a machine-independent manner.

Methods inherited from class java.io.FilterOutputStream

close, write

Methods inherited from class java.lang.Object

clone, equals, finalize, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

Methods inherited from interface java.io.DataOutput

write

Field Detail

written

```
protected int written
```

The number of bytes written to the data output stream so far. If this counter overflows, it will be wrapped to Integer.MAX_VALUE.

Constructor Detail

DataOutputStream

```
public DataOutputStream(OutputStream out)
```

Creates a new data output stream to write data to the specified underlying output stream. The counter `written` is set to zero.

Parameters:

`out - the underlying output stream, to be saved for later use.`

See Also:

FilterOutputStream.out

Method Detail

write

```
public void write(int b)
           throws IOException
```

Writes the specified byte (the low eight bits of the argument `b`) to the underlying output stream. If no exception is thrown, the counter `written` is incremented by 1.

Implements the `write` method of `OutputStream`.

Specified by:

`write` in interface `DataOutput`

Overrides:

`write` in class `FilterOutputStream`

Parameters:

`b` - the byte to be written.

Throws:

`IOException` - if an I/O error occurs.

See Also:

`FilterOutputStream.out`

write

```
public void write(byte[] b,
                  int off,
                  int len)
           throws IOException
```

Writes `len` bytes from the specified byte array starting at offset `off` to the underlying output stream. If no exception is thrown, the counter `written` is incremented by `len`.

Specified by:

`write` in interface `DataOutput`

Overrides:

`write` in class `FilterOutputStream`

Parameters:

b - the data.

off - the start offset in the data.

len - the number of bytes to write.

Throws:

IOException - if an I/O error occurs.

See Also:

FilterOutputStream.out

flush

```
public void flush()
           throws IOException
```

Flushes this data output stream. This forces any buffered output bytes to be written out to the stream.

The `flush` method of `DataOutputStream` calls the `flush` method of its underlying output stream.

Specified by:

flush in interface Flushable

Overrides:

flush in class FilterOutputStream

Throws:

IOException - if an I/O error occurs.

See Also:

FilterOutputStream.out, OutputStream.flush()

writeBoolean

```
public final void writeBoolean(boolean v)
                        throws IOException
```

Writes a `boolean` to the underlying output stream as a 1-byte value. The value `true` is written out as the value `(byte)1`; the value `false` is written out as the value `(byte)0`. If no exception is thrown, the counter `written` is incremented by `1`.

Specified by:

writeBoolean in interface DataOutput

Parameters:

v - a boolean value to be written.

Throws:

IOException - if an I/O error occurs.

See Also:

FilterOutputStream.out

writeByte

```
public final void writeByte(int v)
                     throws IOException
```

Writes out a byte to the underlying output stream as a 1-byte value. If no exception is thrown, the counter written is incremented by 1.

Specified by:

writeByte in interface DataOutput

Parameters:

v - a byte value to be written.

Throws:

IOException - if an I/O error occurs.

See Also:

FilterOutputStream.out

writeShort

```
public final void writeShort(int v)
                      throws IOException
```

Writes a short to the underlying output stream as two bytes, high byte first. If no exception is thrown, the counter written is incremented by 2.

Specified by:

writeShort in interface DataOutput

Parameters:

v - a short to be written.

Throws:

IOException - if an I/O error occurs.

See Also:

FilterOutputStream.out

writeChar

```
public final void writeChar(int v)
                     throws IOException
```

Writes a `char` to the underlying output stream as a 2-byte value, high byte first. If no exception is thrown, the counter `written` is incremented by 2.

Specified by:

`writeChar` in interface `DataOutput`

Parameters:

`v` - a `char` value to be written.

Throws:

`IOException` - if an I/O error occurs.

See Also:

`FilterOutputStream.out`

writeInt

```
public final void writeInt(int v)
                    throws IOException
```

Writes an `int` to the underlying output stream as four bytes, high byte first. If no exception is thrown, the counter `written` is incremented by 4.

Specified by:

`writeInt` in interface `DataOutput`

Parameters:

`v` - an `int` to be written.

Throws:

`IOException` - if an I/O error occurs.

See Also:

`FilterOutputStream.out`

writeLong

```
public final void writeLong(long v)
                     throws IOException
```

Writes a `long` to the underlying output stream as eight bytes, high byte first. In no exception is thrown, the counter `written` is incremented by 8.

Specified by:

writeLong in interface DataOutput

Parameters:

v - a long to be written.

Throws:

IOException - if an I/O error occurs.

See Also:

FilterOutputStream.out

writeFloat

```
public final void writeFloat(float v)
                      throws IOException
```

Converts the float argument to an int using the floatToIntBits method in class Float, and then writes that int value to the underlying output stream as a 4-byte quantity, high byte first. If no exception is thrown, the counter written is incremented by 4.

Specified by:

writeFloat in interface DataOutput

Parameters:

v - a float value to be written.

Throws:

IOException - if an I/O error occurs.

See Also:

FilterOutputStream.out, Float.floatToIntBits(float)

writeDouble

```
public final void writeDouble(double v)
                       throws IOException
```

Converts the double argument to a long using the doubleToLongBits method in class Double, and then writes that long value to the underlying output stream as an 8-byte quantity, high byte first. If no exception is thrown, the counter written is incremented by 8.

Specified by:

writeDouble in interface DataOutput

Parameters:

v - a double value to be written.

Throws:

IOException - if an I/O error occurs.

See Also:

FilterOutputStream.out, Double.doubleToLongBits(double)

writeBytes

```
public final void writeBytes(String s)
                      throws IOException
```

Writes out the string to the underlying output stream as a sequence of bytes. Each character in the string is written out, in sequence, by discarding its high eight bits. If no exception is thrown, the counter `written` is incremented by the length of `s`.

Specified by:

writeBytes in interface DataOutput

Parameters:

s - a string of bytes to be written.

Throws:

IOException - if an I/O error occurs.

See Also:

FilterOutputStream.out

writeChars

```
public final void writeChars(String s)
                      throws IOException
```

Writes a string to the underlying output stream as a sequence of characters. Each character is written to the data output stream as if by the `writeChar` method. If no exception is thrown, the counter `written` is incremented by twice the length of `s`.

Specified by:

writeChars in interface DataOutput

Parameters:

s - a String value to be written.

Throws:

IOException - if an I/O error occurs.

See Also:

writeChar(int), FilterOutputStream.out

writeUTF

```
public final void writeUTF(String str)
                    throws IOException
```

Writes a string to the underlying output stream using modified UTF-8 encoding in a machine-independent manner.

First, two bytes are written to the output stream as if by the `writeShort` method giving the number of bytes to follow. This value is the number of bytes actually written out, not the length of the string. Following the length, each character of the string is output, in sequence, using the modified UTF-8 encoding for the character. If no exception is thrown, the counter `written` is incremented by the total number of bytes written to the output stream. This will be at least two plus the length of `str`, and at most two plus thrice the length of `str`.

Specified by:

`writeUTF` in interface `DataOutput`

Parameters:

`str` - a string to be written.

Throws:

`IOException` - if an I/O error occurs.

size

```
public final int size()
```

Returns the current value of the counter `written`, the number of bytes written to this data output stream so far. If the counter overflows, it will be wrapped to Integer.MAX_VALUE.

Returns:

the value of the written field.

See Also:

```
written
```


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